

Combining Technical Indicators and Equal-Budget Metaheuristic Optimization for Short-Term Mini-Index Futures Prediction

Abstract

Short-horizon trend classification in futures markets is vulnerable to non-stationarity, weak signal-to-noise ratios, and temporal information leakage. An earlier conference precursor combined technical indicators, Information Gain feature selection, and genetic-algorithm tuning, but used randomized cross-validation, a single metaheuristic, and no economic assessment. This study advances that research line through an auditable benchmark on 15,057 five-minute B3 Mini-Index futures bars. Four model backbones (multilayer perceptron, random forest, support vector machine, and 1D convolutional neural network) are optimized by random search, genetic algorithm, particle swarm optimization, differential evolution, and grey wolf optimization under an identical 1,000-evaluation budget. A chronological 60/20/20 train/validation/test partition prevents look-ahead leakage. The study contrasts two validation objectives: a compound $0.60 \text{ MCC} + 0.40 F_1$ fitness and a weighted training/validation accuracy fitness. The common, locked test block is used only after selection for predictive and trading evaluation. Results are reported transparently on the three seeds available in every model-optimizer cell. The direct fitness comparison is descriptive; complementary Friedman and Holm-adjusted Wilcoxon analyses compare models within each protocol across matched optimizer-seed blocks. The study provides a reproducible design for separating the selection criterion from the reported metric and for jointly examining predictive and economic consequences of metaheuristic tuning.

Keywords: Financial time series · Metaheuristic optimization · Hyperparameter tuning · Matthews correlation coefficient · Temporal validation · Intraday futures

1 Introduction

Short-term price forecasting in high-frequency derivatives markets represents one of the most computationally demanding applications of neural computing [1, 2]. Among traded financial instruments, futures contracts are particularly sought after by institutional and retail market participants due to their high liquidity, leverage potential, and utility for both risk hedging against price fluctuations and speculative trading

strategies [3, 4]. These contracts establish an agreement to purchase or sell an underlying asset at a predetermined price on a future date, rendering them exceptionally sensitive to order flow dynamics, short-term market expectations, and macroeconomic announcements [5, 6]. In Brazil, the Mini-Index futures contract (symbol: WIN) on Brasil Bolsa Balcão (B3), referenced to the Ibovespa stock index, exhibits immense daily trading volume and constitutes a prime domain for computational intelligence research in financial time series [1, 7]. However, price series derived from 5-minute intraday candles exhibit severe non-stationarity, low signal-to-noise ratios, structural regime shifts, and rapid dissipation of arbitrage patterns as market participants exploit temporary inefficiencies [8, 9]. These characteristics render accurate directional trend classification a methodologically challenging task that demands both robust feature engineering and principled hyperparameter optimization [10].

Early attempts to formalize market patterns and predict directional movements relied on static linear models, such as autoregressive integrated moving average (ARIMA) techniques [11]. As finance and computer science converged under the computational intelligence paradigm [12], data-driven machine learning (ML) models gained widespread adoption for financial classification tasks. Supervised learning algorithms—including Random Forests (RF) [13], Support Vector Machines (SVM) [14], Multilayer Perceptrons (MLP) [15], and 1D Convolutional Neural Networks (1D-CNN) [16]—have demonstrated strong capabilities in learning complex, non-linear decision boundaries from historical market features. A central prerequisite for building effective classifiers involves technical analysis, wherein mathematical transformations of price and volume series—such as Simple Moving Averages (SMA), Exponential Moving Averages (EMA), and standard deviation envelopes—are computed to capture trend momentum and volatility regimes [10, 17]. Nevertheless, computing high-dimensional pools of technical indicators introduces significant redundancy and noise, creating a feature space that degrades classifier generalization and increases computational overhead [18, 19]. Consequently, applying feature selection techniques to extract compact, highly informative indicator subsets becomes vital for stabilizing training dynamics and improving out-of-sample robustness [3, 20].

Beyond feature engineering and selection, classifier performance depends critically on hyperparameter configuration [21]. In neural network architectures such as MLPs and 1D-CNNs, structural parameters (e.g., hidden layer width, filter counts), regularization coefficients (L2 weight decay α , dropout rate p_{drop}), and training settings (learning rate lr , mini-batch size B) jointly define a complex, non-convex optimization surface over a mixed discrete-continuous search space $\theta \in \mathbb{R}^D$ [9, 22]. Traditional gradient-based algorithms such as Stochastic Gradient Descent or Adam operate effectively on differentiable weight-loss surfaces but cannot directly navigate the non-differentiable, combinatorial hyperparameter landscape [23, 24]. To overcome these limitations, metaheuristic optimization algorithms (MOAs)—spanning evolutionary algorithms such as Genetic Algorithms (GA) [25] and Differential Evolution (DE) [26], as well as swarm intelligence paradigms such as Particle Swarm Optimization (PSO) [27] and Grey Wolf Optimizer (GWO) [23]—have gained substantial prominence for automated hyperparameter tuning in neural computing applications [1, 4]. These population-based methods maintain diverse candidate pools and employ stochastic

exploration-exploitation operators, enabling effective traversal of rugged, multi-modal fitness landscapes without requiring gradient information [23].

Despite the growing literature on metaheuristic-tuned financial classifiers published in premier neural computing journals [1, 4, 9, 16], existing financial machine learning studies overwhelmingly suffer from three major methodological flaws:

1. **The Budget Inequality Flaw:** Benchmark studies frequently evaluate metaheuristics under unconstrained or unequal evaluation counts (e.g., comparing a GA running for 100 generations with a PSO running for 30 iterations) [9]. Because stochastic search performance scales monotonically with candidate evaluation count, reported performance superiority in prior studies is often an artifact of unequal evaluation budgets rather than inherent algorithmic efficiency.
2. **The Temporal Data Leakage Flaw:** Standard randomized K -fold cross-validation or randomized data shuffling is routinely applied to time series data [8, 9]. Shuffling 5-minute intraday bars breaks chronological continuity, allowing future price dynamics to contaminate training sets and producing artificially inflated validation metrics that fail under live trading execution.
3. **The Classification-Economic Disconnect Flaw:** Models are overwhelmingly tuned and evaluated solely on raw classification accuracy or cross-entropy loss [28]. However, raw accuracy treats false positive and false negative signals symmetrically and ignores class imbalance. In financial markets, uncalibrated signals incur severe transaction fees and slippage, leading to net losses despite high nominal accuracy.

To resolve these methodological limitations, this paper presents a controlled multi-optimizer benchmark framework for short-term neural intraday trend classification on 5-minute B3 Mini-Index futures data. Incorporating Information Gain feature selection to extract a compact 7-indicator subset from 66 raw technical indicators, we systematically evaluate five distinct optimizer families—Random Search (RS), Genetic Algorithm (GA), Particle Swarm Optimization (PSO), Differential Evolution (DE), and Grey Wolf Optimizer (GWO)—across four diverse classifier backbones (MLP, SVM, RF, 1D-CNN). All optimizers operate under a strictly equal candidate evaluation budget cap of $N_{\text{eval}} = 1,000$ evaluations per seed within a leakage-free 60/20/20 chronological split, contrasting a compound MCC-driven objective ($0.60\text{MCC} + 0.40F_1$) against weighted accuracy fitness. Through non-parametric Friedman and post-hoc Wilcoxon–Holm analyses, we examine optimizer performance across matched blocks and validate configurations via out-of-sample financial backtesting under transaction costs.

The remainder of this manuscript is organized as follows: Section 2 presents a systematic review of the literature, establishing current state-of-the-art paradigms and critical research gaps. Section 3 details the dataset, Information Gain feature selection, model backbones, search-space formulations, metaheuristic operators, and evaluation protocol. Section 4 presents the empirical benchmark results, convergence dynamics, statistical analyses, and financial backtest outcomes. Section 5 interprets search dynamics, trade-offs, and practical deployment implications. Finally, Section 6 concludes the paper and outlines future research directions.

2 Systematic Review of Literature

Predicting short-term directional price movements in high-frequency financial markets remains one of the most demanding applications of neural computing due to low signal-to-noise ratios, rapid regime shifts, and severe non-stationarity [2, 29]. Supervised machine learning algorithms and technical indicator representations have been widely deployed to capture non-linear intraday patterns; however, predictive efficacy depends heavily on input feature selection, hyperparameter tuning, and validation protocol integrity [18, 30]. This section presents a structured Systematic Review of Literature (SRL) examining prior work across input representations, metaheuristic optimization paradigms, and validation designs, synthesizing current state-of-the-art contributions and formalizing the critical methodological gaps that motivate our benchmark framework.

2.1 Literature Selection Protocol and Research Questions

To ensure methodological rigor and eliminate reporting bias, relevant literature was retrieved following an explicit PRISMA-guided protocol across major scholarly databases: IEEE Xplore, Scopus, Web of Science, and SpringerLink (specifically targeting *Neural Computing and Applications*). The search strategy combined boolean queries for optimization paradigms ("metaheuristic", "genetic algorithm", "particle swarm", "differential evolution", "grey wolf optimization"), machine learning models ("multilayer perceptron", "random forest", "support vector machine", "1D-CNN"), and financial application domains ("futures market", "intraday forecasting", "technical indicators").

The review was bounded to peer-reviewed journal articles and top-tier conference proceedings published between 2015 and 2026. Inclusion criteria mandated that candidate studies: (i) evaluate automated feature selection or hyperparameter tuning on financial or complex time-series classification/forecasting, (ii) specify validation protocols and search spaces, and (iii) report empirical predictive or financial metrics. Studies lacking reproducible experimental parameters or failing to disclose optimization constraints were excluded. The systematic review is organized around three central Research Questions (RQs):

1. **RQ1 (Budget Parity)**: How do stochastic metaheuristic search operators (RS, GA, PSO, DE, GWO) compare across distinct classifier backbones (RF, SVM, MLP, 1D-CNN) when restricted to a strictly equal candidate evaluation budget cap ($N_{\text{eval}} = 1,000$)?
2. **RQ2 (Temporal Leakage)**: What is the quantitative impact on out-of-sample generalization when comparing chronological sequential temporal splits against randomized cross-validation on high-frequency financial time series?
3. **RQ3 (Economic Alignment)**: How effectively do classification-based validation fitness objectives (e.g., compound MCC/ F_1 vs. weighted accuracy) align with out-of-sample intraday trading metrics (Net Return %, Sharpe ratio, Maximum Drawdown) under realistic exchange transaction costs?

2.2 Classifier Backbones and Input Representations

Supervised intraday trend prediction relies heavily on the quality and information density of input features [18]. Raw high-frequency bar data (OHLCV) exhibit extreme microstructural noise and high variance. Consequently, domain-specific technical analysis indicators—spanning momentum indicators, moving averages, and volatility envelopes—are widely computed to extract directional signals [10, 17, 31]. However, calculating dozens of technical indicators across multiple lookback windows introduces severe multicollinearity and high dimensionality, which can destabilize model training and induce overfitting [19, 20].

Information-theoretic feature selection provides a principled approach to reduce feature space dimensionality. Information Gain (IG) attribute filtering measures the mutual information between individual indicators and the target trend label, ranking features according to their predictive entropy reduction [3, 19]. Filtering out redundant or noisy indicators isolates a compact, highly informative feature core, which stabilizes fitted classification boundaries and lowers computational overhead during downstream hyperparameter search [20].

Downstream predictive backbones span diverse algorithmic hypothesis spaces:

- **Random Forest (RF)**: An ensemble of decision trees constructed via bootstrap aggregation (bagging) and random feature subset selection, highly resilient to feature noise and non-linear interactions [13].
- **Support Vector Machines (SVM)**: Convex maximum-margin classifiers that map non-linearly separable indicator subsets into high-dimensional feature spaces using kernel functions (e.g., Radial Basis Function) [14].
- **Multilayer Perceptron (MLP)**: Fully-connected feedforward neural networks that learn complex non-linear decision maps via error backpropagation, sensitive to layer widths, learning rates, and regularization settings [15].
- **1D Convolutional Neural Network (1D-CNN)**: Deep learning architectures employing 1D temporal convolution filters and pooling layers to extract local spatial-temporal patterns directly from sequential indicator windows [9, 16].

Prior research confirms that classifier ranking is strongly dependent on input feature representation, demonstrating that feature subset selection and model hyperparameter configuration must be evaluated jointly [13, 32].

2.3 Metaheuristic Optimization Paradigms

Hyperparameter optimization is critical to prevent underfitting or severe overfitting in financial learning models. Classical gradient-based optimization methods (e.g., SGD, Adam) require smooth, differentiable loss surfaces with respect to weights, but cannot navigate discrete structural parameters (e.g., hidden neurons, filter counts, batch sizes) or non-differentiable regularization settings [9, 22]. Population-based metaheuristic algorithms offer derivative-free global search capabilities over rugged, non-convex, mixed discrete-continuous search spaces $\theta \in \mathbb{R}^D$ [23, 24].

Four primary population-based metaheuristic families dominate the financial ML literature:

- **Genetic Algorithms (GA)**: Evolutionary optimization schemes simulating natural selection through population selection, uniform/single-point crossover, and mutation operators to evolve candidate solution vectors [21, 25].
- **Particle Swarm Optimization (PSO)**: Swarm intelligence algorithms simulating avian flocking, where individual particles dynamically update velocity vectors based on personal historical bests and global swarm best positions [27, 33].
- **Differential Evolution (DE)**: Stochastic population-based vector search methods that generate trial vectors using vector differences for self-referential mutation and crossover operations [26].
- **Grey Wolf Optimizer (GWO)**: Bio-inspired metaheuristics modeling the hunting social hierarchy ($\alpha, \beta, \delta, \omega$) and prey encirclement mechanisms of grey wolves [4, 23].

Random Search (RS) provides an essential baseline sampler in metaheuristic benchmarks, generating uniform random configurations across bounded parameter spaces to test whether guided search operators yield true efficiency gains beyond unguided sampling [34].

While mathematical update formulations for these optimizers are detailed in Section 3, existing literature frequently compares these algorithms under unconstrained or unequal candidate evaluation counts (e.g., running GA for 100 generations vs. PSO for 30 iterations without matching population sizes). Such unequal setups obscure whether reported performance gains stem from superior search operators or simply higher computational evaluation budgets [9, 15].

2.4 Comparative Synthesis of Representative Studies

To establish the state of the art and pinpoint persistent methodological gaps, Table 1 synthesizes 15 representative published studies from 2015 to 2026 alongside the present benchmark work. Studies are evaluated across six technical dimensions: Study & Year, Application Domain & Data, Model Backbones, Optimization & Feature Paradigms, Validation Protocol, and Identified Methodological Gap.

2.5 Critical Methodological Gaps and Research Motivation

As synthesized in Table 1, three primary methodological flaws permeate the existing literature on metaheuristic financial forecasting:

1. *The Budget Inequality Flaw*: Prior studies routinely compare metaheuristics without enforcing a strict candidate evaluation cap (N_{eval}). Because stochastic optimization performance scales with function evaluations, superiority claims in unconstrained studies often reflect unequal computational budgets rather than algorithmic search efficiency [9, 15].
2. *The Temporal Data Leakage Flaw*: Financial machine learning research frequently relies on randomized K -fold cross-validation or data shuffling [8, 37]. Shuffling sequential intraday price bars allows future market information to contaminate historical training folds, yielding artificially inflated predictive accuracy scores that collapse during live execution.

Table 1 Systematic Literature Review comparison matrix auditing representative studies in financial machine learning, metaheuristic hyperparameter tuning, and temporal validation protocols against identified methodological gaps.

Study & Year	Domain & Data	Models	Optimization & Features	Validation Protocol	Methodological Gap
Ballings et al. (2015) [13] Ding et al. (2011) [21]	Equities (5,767 firms) Stock index trend	SVM, RF, AdaBoost Neural Networks	Grid search / Manual Hybrid GA-BP	Cross-validation Standard split	Absence of metaheuristic search. Unequal evaluation budget across optimizers.
Naik & Mohan (2019) [14] Ecer et al. (2020) [15]	Equities (NSE India) Financial trend	ANN classifier MLP classifier	Boruta feature selection GA and PSO tuning	Train/test split K -fold CV	Static parameters; no metaheuristics. Unconstrained budget; single classifier.
Billah et al. (2024) [10]	Intraday equities	Heuristic models	Moving average rules	Standard split	Absence of neural hyperparameter tuning.
Bareket & Párv (2024) [35] Bandgar et al. (2025) [16] Rahim et al. (2025) [36] Dhingra et al. (2025) [9]	Surge detection Financial forecasting 5-min futures Solar PV forecasting	Adaptive ML Ensembles Single neural model LSTM, CNN-LSTM	Technical indicators Default tuning Single GA optimization RAdam, Lookahead, Adam	Medium-term split Random split Train/test split Sequential split	Evaluation restricted to static features. No evaluation budget cap; data leakage. Single GA; no multi-swarm benchmark. Non-financial domain; stochastic optimizers omitted.
Mutinda & Yong (2026) [32] Simatupang et al. (2026) [37]	Financial market Stock index (IDX)	Ensembles Bayesian RF	SEDT feature selection Bayesian optimization	K -fold CV Random K -fold CV	Feature selection without tuning. Random CV causes temporal data leakage.
Al Bannoud et al. (2026) [4]	Kinetics time series	ANN-ODE surrogate	GWO, PSO, GA, ABC	Multi-partition	Non-financial; unconstrained search budget.
Bashir et al. (2026) [1] [Anonymized Precursor A] (2025) [3] [Anonymized Precursor B] (2026) [38] Present Work	Equities (NSE India) Intraday B3 futures Intraday B3 futures Intraday B3 futures Intraday B3 futures	Bagging Ensemble Random Forest MLP, SVM, RF MLP, SVM, RF, 1D-CNN	(1+1) Evolutionary Algo InfoGain + Default RF InfoGain + Single GA InfoGain + RS, GA, PSO, DE, GWO	Train/test split Sequential 60/20/20 5-fold CV Sequential 60/20/20	Single EA evaluated without budget cap. Untuned baseline classifier without metaheuristics. Single GA; unconstrained 1,000-generation budget. Equal budget ($N_{\text{eval}} = 1,000$), 3 seeds, fee backtest.

3. *The Classification-Economic Disconnect Flaw:* Hyperparameter tuning is overwhelmingly conducted against raw accuracy or cross-entropy loss [3, 28]. In high-frequency futures trading, raw accuracy ignores class imbalance and trade execution costs. A classifier achieving high nominal accuracy can incur net financial losses once exchange transaction fees and slippage are deducted.

To address these limitations, this paper establishes an auditable multi-optimizer benchmark framework on 15,057 five-minute B3 Mini-Index futures bars. Leveraging Information Gain feature selection to extract a compact 7-indicator subset, we systematically evaluate five optimizer families (RS, GA, PSO, DE, GWO) across four diverse model backbones (MLP, SVM, RF, 1D-CNN). All optimizers operate under a strictly locked candidate evaluation cap of $N_{\text{eval}} = 1,000$ evaluations per seed across three common random seeds within a leakage-free 60/20/20 chronological partition. Furthermore, we contrast two distinct validation fitness objectives (a compound

0.60 MCC + 0.40 F_1 objective vs. weighted accuracy) and evaluate held-out trading profitability under realistic exchange transaction costs.

3 Materials and Methods

3.1 Dataset, Target, and Feature Representation

The empirical dataset contains 15,057 clean, chronologically ordered five-minute bars for B3 Mini-Index futures (WIN), drawn from BovDB resources [6, 7]. The binary target is the next-bar directional movement, with $y_k = 1$ for an uptrend and $y_k = 0$ otherwise. The source pipeline derives 66 price-, moving-average-, volatility-, volume-, and trade-frequency descriptors. In keeping with the precursor, Information Gain selects seven descriptors: two EMA differences, their normalized counterparts, normalized SMA₉, EMA₉₋₃, and normalized EMA₇. This preserves the predecessor’s compact feature representation while changing the validation and optimizer design.

3.2 Temporal Protocol and Shared Search Controls

All runs follow the same chronological 60/20/20 partition: the first 9,034 bars form the training block, the following 3,011 bars form the validation block, and the final 3,012 bars form the locked test block. No shuffling is applied. The test block is never used to rank candidates or select a fitness function.

Four classifier backbones are assessed: MLP, Random Forest, SVM, and 1D-CNN. Their hyperparameters cover architecture size, regularization, learning rate, dropout, batch size, tree settings, and kernel settings. Each backbone is searched by RS, GA, PSO, DE, and GWO. To make their comparisons interpretable, every optimizer receives the same candidate-evaluation allowance:

$$N_{\text{eval}} = 1,000 \quad \text{evaluations per optimizer and seed.} \quad (1)$$

The optimizers retain their standard search roles: RS samples feasible configurations; GA uses tournament selection, crossover, and mutation; PSO uses cognitive and social attraction; DE uses differential perturbation and crossover; and GWO updates a population through its α , β , and δ leaders. The purpose is not to tune their internal operators asymmetrically, but to compare their model-selection behaviour under the same evaluation opportunity.

3.3 Dual-Fitness Design

Figure 1 depicts the controlled comparison. Protocol A ranks candidates on the validation block by a compound imbalance-aware fitness,

$$f_{\text{MCC}/F_1}(\boldsymbol{\theta}) = 0.60 \text{MCC}_{\text{val}}(\boldsymbol{\theta}) + 0.40 F_{1,\text{val}}(\boldsymbol{\theta}), \quad (2)$$

whereas Protocol B ranks candidates by a weighted training/validation accuracy,

$$f_{\text{Acc}}(\boldsymbol{\theta}) = 0.40 \text{Accuracy}_{\text{train}}(\boldsymbol{\theta}) + 0.60 \text{Accuracy}_{\text{val}}(\boldsymbol{\theta}). \quad (3)$$

The protocols differ only in this validation objective. Data, features, model families, optimizer families, budget, and test block are shared. After selection, both protocols report test MCC, test accuracy, and test F_1 .

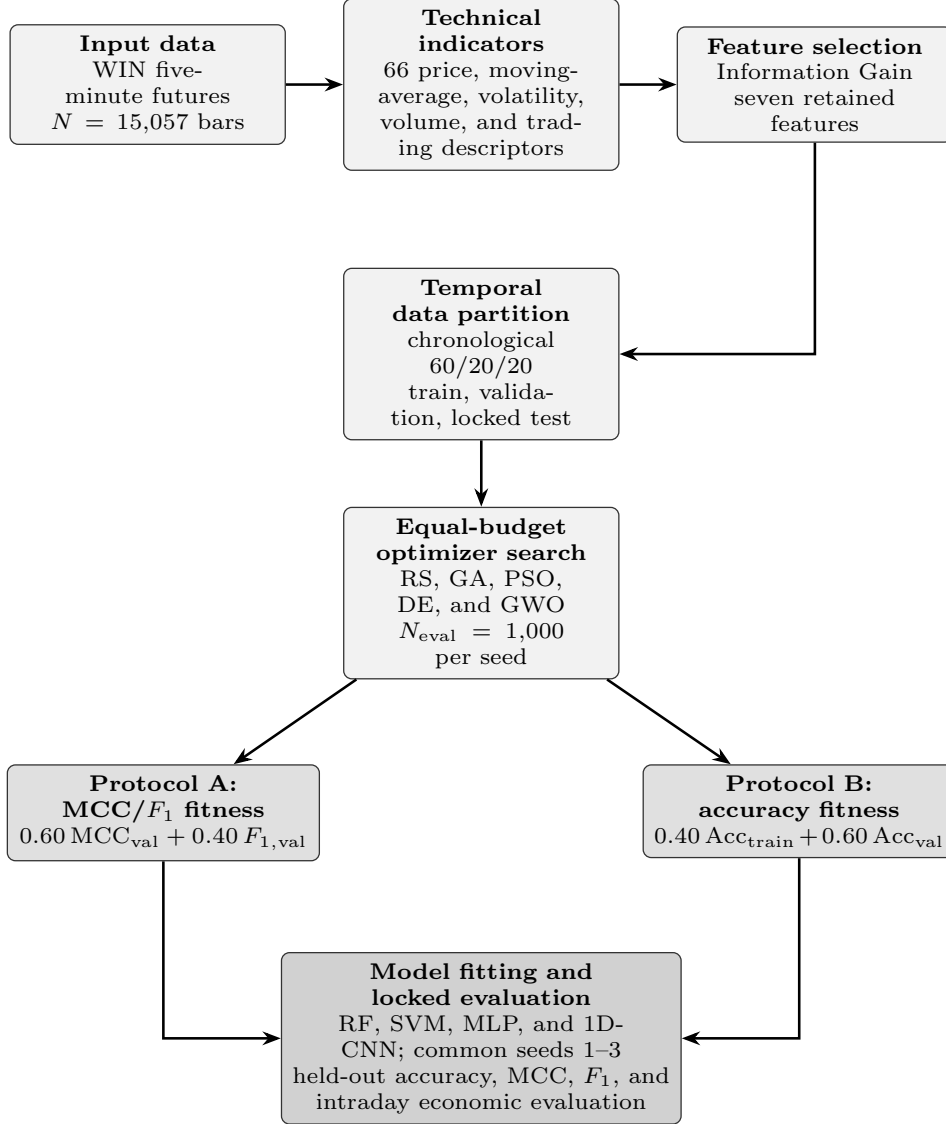


Fig. 1 Overview of the proposed methodology. The pipeline retains the precursor’s sequence of intraday data preparation, indicator construction, feature selection, model fitting, and final evaluation; the new benchmark inserts a chronological partition, equal-budget optimizer comparison, and two explicitly separated validation objectives.

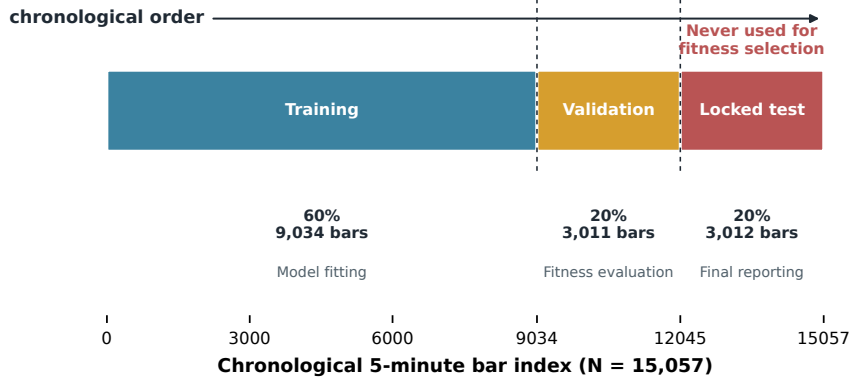


Fig. 2 Chronological 60/20/20 partition. The final test block is unavailable during feature fitting, candidate ranking, and hyperparameter selection.

3.4 Evidence Scope, Statistical Analysis, and Economic Evaluation

Three seeds (1–3) are available for every model–optimizer combination in both protocols. Therefore, the direct protocol contrast uses 20 cell means per protocol, each averaged over the same three seeds, and is presented descriptively. It is not a 20-seed confirmatory test of one fitness against another.

A complementary model-level analysis is performed within each holdout protocol. The five optimizer–seed combinations form 15 matched blocks for the four model backbones. Friedman tests assess global rank differences, followed by paired Wilcoxon tests with Holm adjustment for post-hoc MLP contrasts. This analysis answers whether models separate within a fixed protocol; it does not infer a global fitness advantage.

Finally, the locked predictions are evaluated through an intraday long/short trading simulation. The analysis reports total profit, profit factor, annualized Sharpe ratio, and maximum drawdown. Economic results are interpreted jointly with predictive metrics because a trading strategy can have different return and drawdown characteristics despite similar classification accuracy.

4 Results

4.1 Predictive Outcomes Under Two Selection Objectives

Figure 3 reports held-out accuracy and MCC for the 20 model–optimizer cells. Each line connects a cell evaluated under the two fitness functions and is averaged over the same three seeds. Across cells, the descriptive macro mean test accuracy is 0.623 for MCC/ F_1 fitness and 0.617 for accuracy fitness. The corresponding macro mean test MCC values are 0.247 and 0.235. These quantities summarize the available cells; they are not population estimates.

The result is informative because it rejects a simplistic expectation that selecting on validation accuracy must increase every held-out accuracy value. Some configurations improve under accuracy fitness, whereas others decline, and the corresponding MCC changes are likewise heterogeneous. The selection objective thus interacts with backbone and optimizer rather than imposing a uniform ordering.

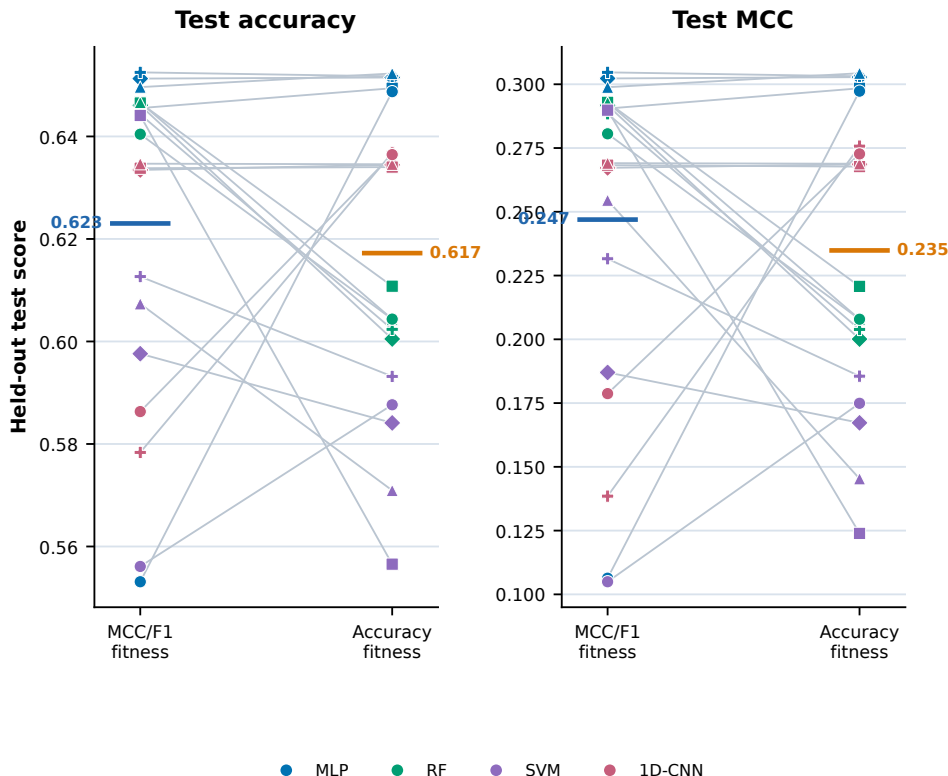
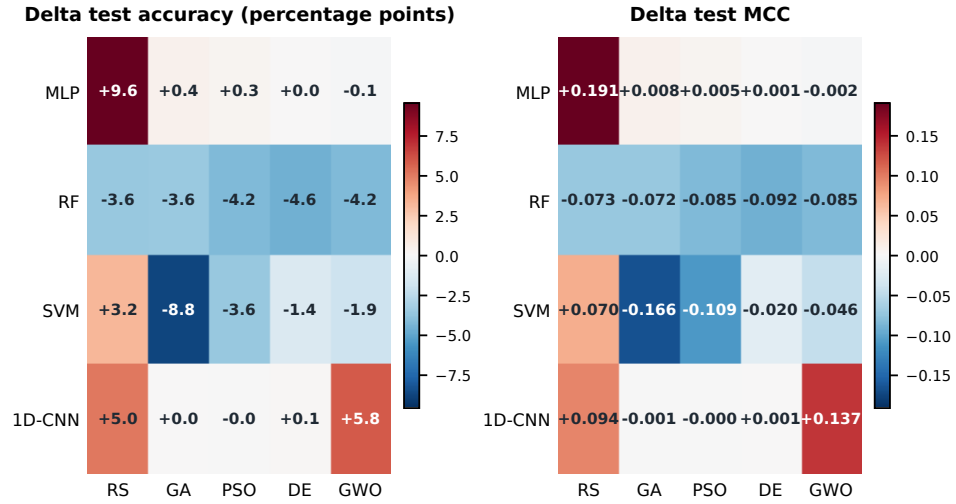


Fig. 3 Held-out test outcomes under the two validation objectives. Each connection is one model-optimizer cell averaged over paired seeds 1–3. Colour denotes the model backbone; marker shape denotes the optimizer. Horizontal segments show macro means across the 20 cells.

Figure 4 localizes the contrast by model and optimizer. Positive cells favour accuracy fitness and negative cells favour compound MCC/F_1 fitness. The heterogeneity reinforces the need to distinguish a selection objective from a reported test metric and motivates a larger common-seed study before making global superiority claims.

4.2 Search Dynamics

Figure 5 uses MLP as a reference backbone to visualize the objective being optimized. The left panel follows compound MCC/F_1 validation fitness, while the right panel follows validation accuracy. The vertical scales differ by construction and must not be interpreted as a direct score comparison. The trajectories show the progress of



Delta = accuracy-fitness protocol minus MCC/F1-fitness protocol; positive values favour accuracy fitness.

Fig. 4 Configuration-level difference between protocols (accuracy fitness minus MCC/ F_1 fitness) on the locked test set. Left: accuracy in percentage points. Right: MCC in raw units.

each optimizer under its assigned objective and make the distinction between the two experiments explicit.

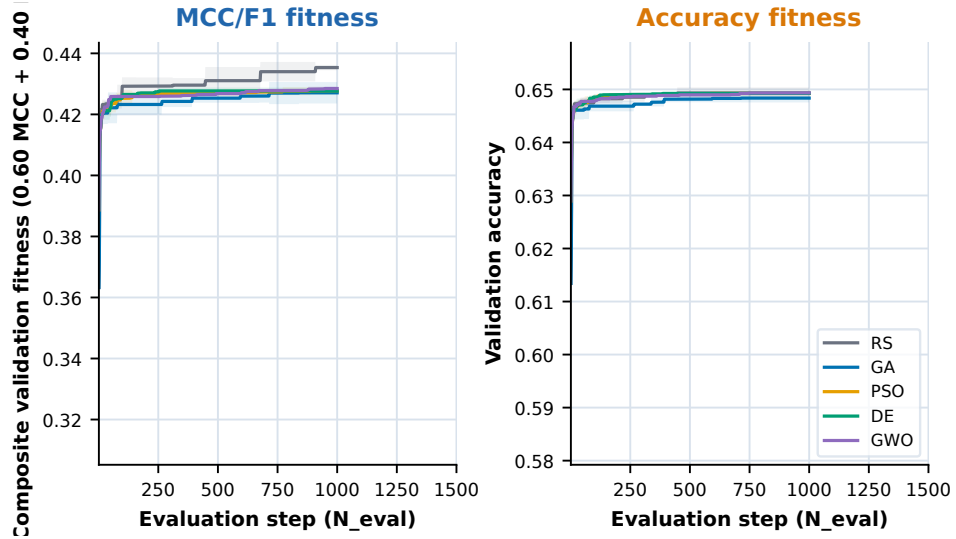


Fig. 5 MLP convergence under the two validation objectives. Lines are means and shaded regions are ± 1 standard deviation across paired seeds 1–3. Left: compound MCC/ F_1 fitness. Right: validation accuracy.

4.3 Exploratory Statistical Evidence

Within Protocol A, the Friedman comparison across 15 matched optimizer–seed blocks identifies differences in test MCC among the four model backbones ($\chi^2 = 22.84$, $p = 4.36 \times 10^{-5}$). Within Protocol B, the analogous test on accuracy is also significant ($\chi^2 = 42.92$, $p = 2.56 \times 10^{-9}$), with MLP receiving mean rank 1.00. In Protocol B, Holm-adjusted MLP contrasts are significant against CNN, RF, and SVM ($p_{\text{Holm}} = 0.00194$ in each comparison).

These tests are exploratory and model-level. They do not test MCC/ F_1 fitness against accuracy fitness, and their 15 blocks should not be confused with 15 independent market periods. Figure 6 makes this scope visible.

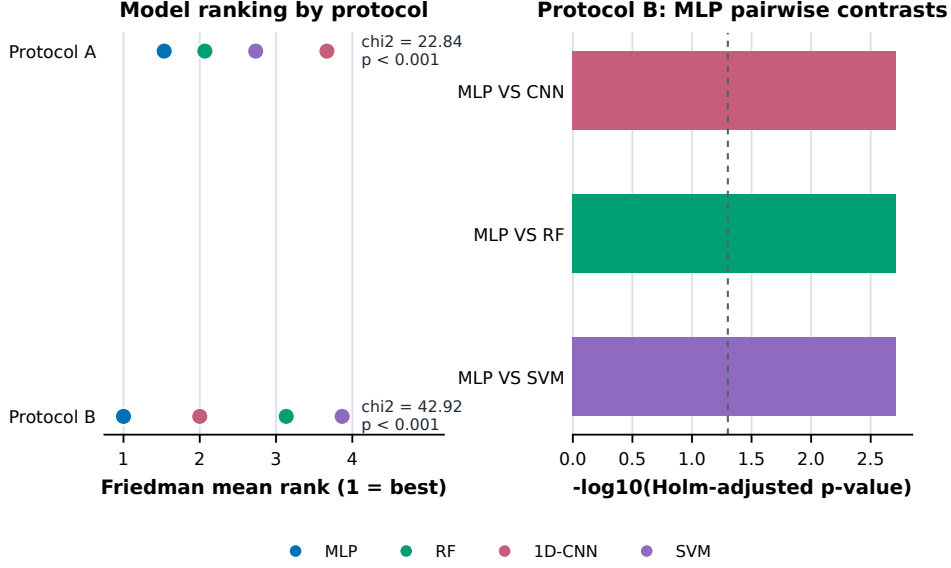


Fig. 6 Exploratory statistical evidence within each holdout protocol. Left: Friedman mean ranks across 15 matched optimizer–seed blocks (lower is better). Right: Holm-adjusted paired Wilcoxon evidence for MLP versus the other models under accuracy fitness; the dashed line denotes $\alpha = 0.05$.

4.4 Economic Evaluation

The economic evaluation reveals a related but distinct ranking. Under MCC/ F_1 fitness, RF has the largest mean total profit (26,465 points), but its Holm-adjusted profit contrasts are not significant. Under accuracy fitness, MLP has the largest mean total profit (27,737 points). MLP’s profit advantage is significant against RF ($p_{\text{Holm}} = 0.00018$) and SVM ($p_{\text{Holm}} = 0.00305$), but not against CNN ($p_{\text{Holm}} = 0.169$). RF has the least negative mean drawdown under accuracy fitness (−1,866 points).

Figure 7 shows why accuracy alone is insufficient for selection: total profit, dispersion, and drawdown do not always identify the same preferred configuration. The

comparison is exploratory but establishes the economic endpoint absent from the precursor study.

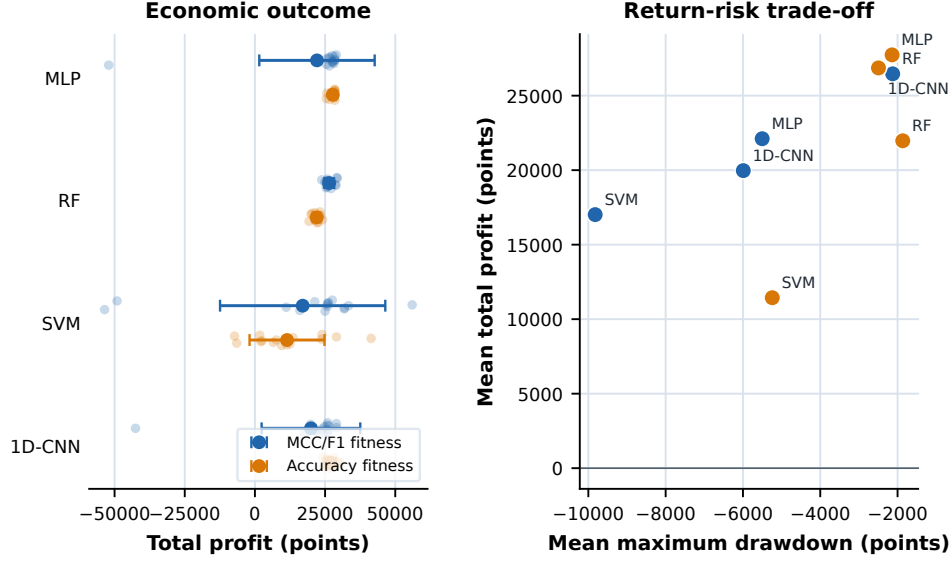


Fig. 7 Exploratory economic evaluation on the locked test block. Left: total-profit observations and mean ± 1 standard deviation across 15 optimizer-seed blocks. Right: mean profit versus mean maximum drawdown; values farther right have less severe drawdown.

5 Discussion

The main advance over the GA-based precursor is methodological control. The earlier work offered a useful proof of concept for combining technical indicators, feature selection, and GA tuning, but it could not distinguish optimizer behaviour from the consequences of randomized validation and unequal search opportunity. The present design makes these assumptions visible: chronological data ordering is preserved, evaluation budgets are fixed, and the validation criterion is stated separately from the final test metrics.

The direct comparison does not support a simple rule that accuracy fitness should dominate when accuracy is reported, or that MCC/ F_1 fitness should dominate every robust metric. The paired cells show positive and negative changes under each criterion. This is plausible in intraday markets, where validation performance is affected by small decision-boundary changes and the final chronological block may represent a different regime. For this reason, the appropriate interpretation is objective-dependent behaviour, not a universal ranking of the two fitness functions.

The within-protocol exploratory analysis provides a more bounded result. MLP ranks first under the accuracy-fitness protocol across matched optimizer-seed blocks, and its adjusted pairwise contrasts against the other backbones are significant. The

economic analysis, however, demonstrates that the predictive ordering does not settle the decision alone: RF attains the smallest mean drawdown under accuracy fitness, whereas MLP obtains the largest mean profit. A deployment decision must therefore specify whether return, downside risk, turnover, or stability is the governing criterion.

Several limitations should guide interpretation. Only three seeds are common to all model-optimizer cells in both protocols. The direct dual-fitness results are therefore descriptive, and the model-level Friedman/Wilcoxon analyses are exploratory. The data cover one liquid Brazilian futures market and one chronological test segment; results may differ across instruments and regimes. Finally, the fixed 1,000-evaluation allowance is appropriate for a controlled benchmark but does not reveal long-budget asymptotic behaviour.

The next study phase is clear: run a preregistered common 20-seed design for every cell, retain the same temporal split and candidate budget, record all validation metrics at each candidate evaluation, and apply paired tests separately to MCC, accuracy, and economic outcomes. That extension would permit confirmatory inference about fitness choice while preserving the methodological strengths established here.

6 Conclusion

This study transforms a GA-based conference precursor into a controlled Neural Computing and Applications benchmark for neural intraday trend classification. It preserves the predecessor’s practical insight—a compact, Information-Gain-selected technical representation can support metaheuristic hyperparameter search—but replaces randomized validation and a single-optimizer design with a chronological dual-fitness protocol, five budget-matched optimizers, and locked predictive and economic assessment.

The current evidence indicates that the choice between compound MCC/ F_1 fitness and accuracy fitness produces configuration-dependent changes in both held-out accuracy and MCC. Within the accuracy-fitness protocol, MLP is the strongest model in the exploratory predictive analysis and leads mean total profit, although RF has the smallest mean drawdown. These results reinforce the central methodological lesson: selection objective, reported metric, and economic utility must be treated as related but distinct quantities.

The manuscript deliberately avoids a confirmatory claim about a universal best fitness function because only three paired seeds are currently available across every cell. Completing the common 20-seed design is the necessary next step. The resulting framework nevertheless provides a transparent and reproducible basis for future optimizer comparisons in non-stationary financial time series.

Declarations

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Conflict of Interest

The authors declare that they have no conflict of interest or competing financial interests.

Ethical Approval

This article does not contain any studies with human participants or animals performed by any of the authors.

Data and Code Availability

The datasets generated and analyzed during the current study, along with the complete source code for all optimization algorithms, benchmark scripts, and evaluation pipelines, are available in the project repository upon reasonable request.

Author Contributions

All authors contributed to the study conception and design. Data collection and experimental execution were performed by [Anonymous for double-blind review]. Statistical analysis and manuscript preparation were carried out by [Anonymous for double-blind review]. All authors read and approved the final manuscript.

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